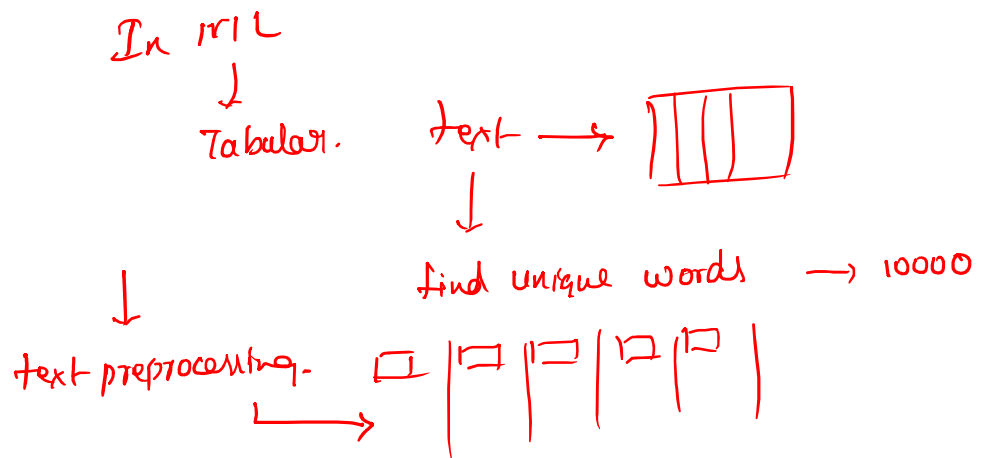
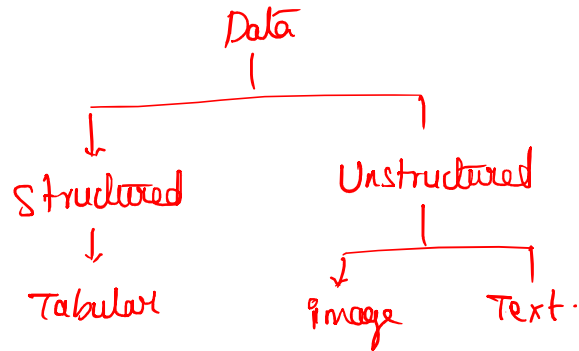


Text Preprocessing:



NLP (Natural Language Processing)

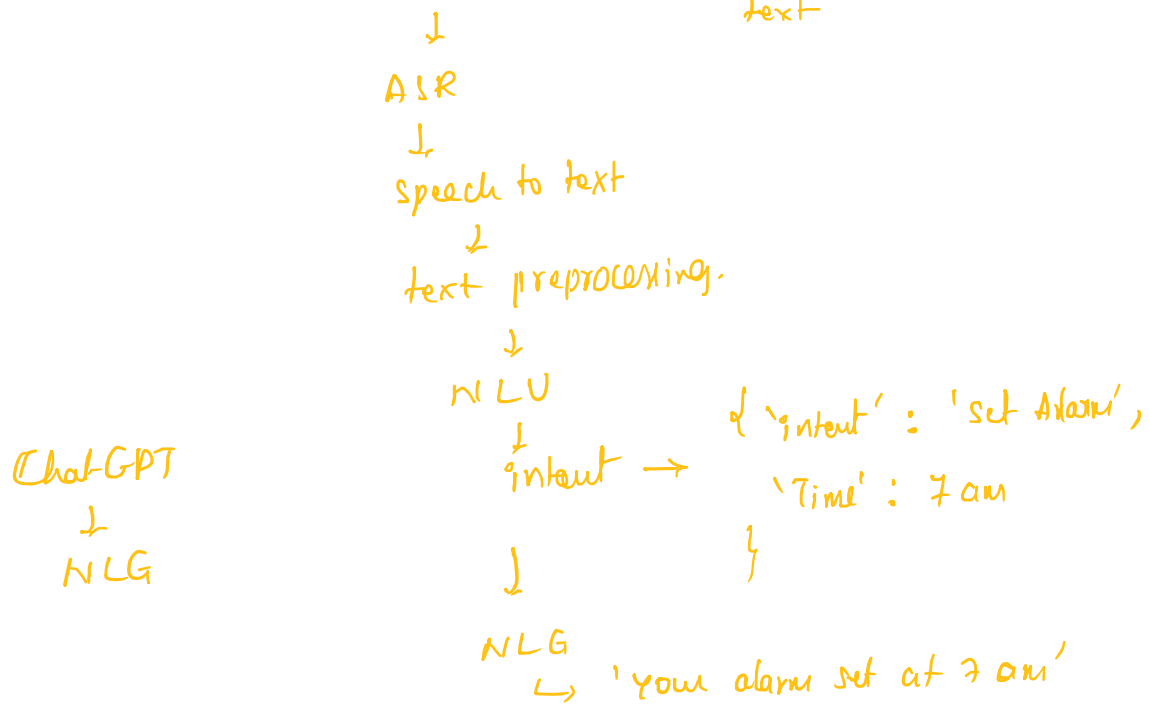
Ram
↓ ↓ ↓

Eng → Letters → ASCII
 ↳ words
 ↳ sentences
 ↳ paragraph

NLP < NLU (Natural Lang Understanding)
 NLG (Natural lang Generator)

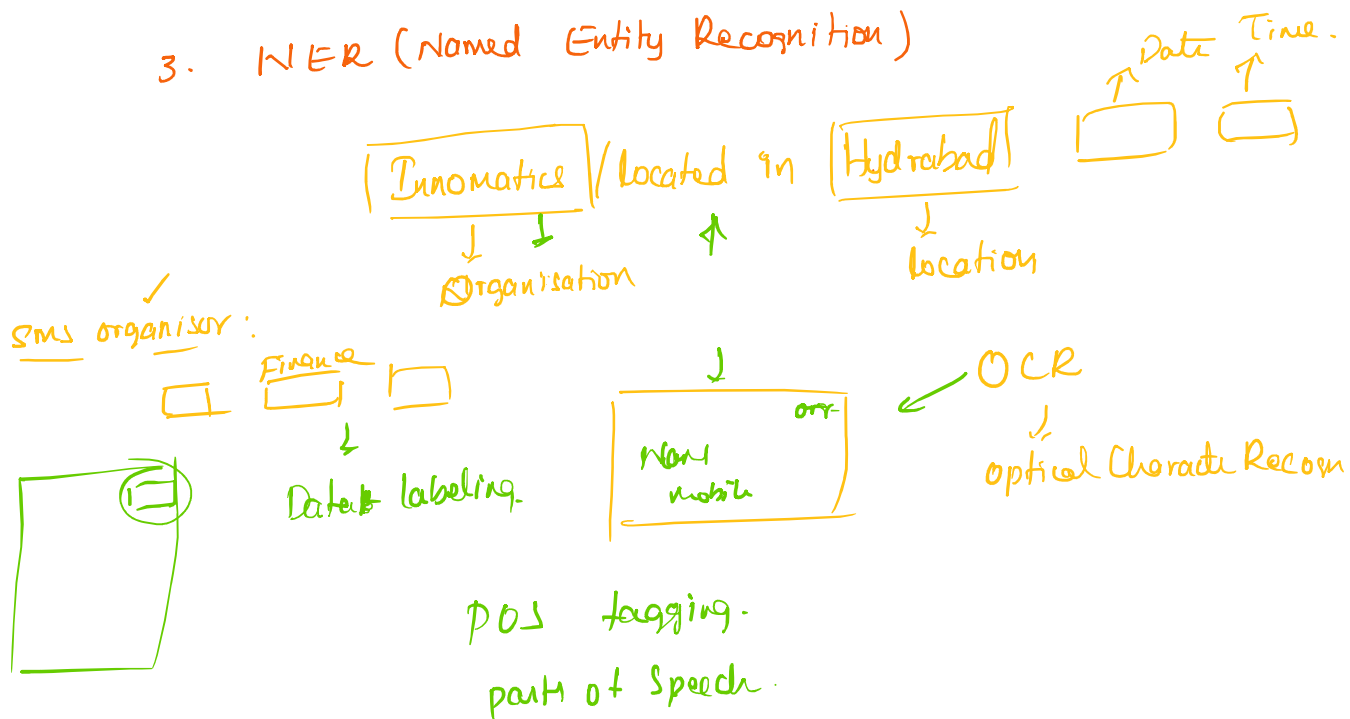
Google Ass:

voice msg: Set Alarm at 7am, text



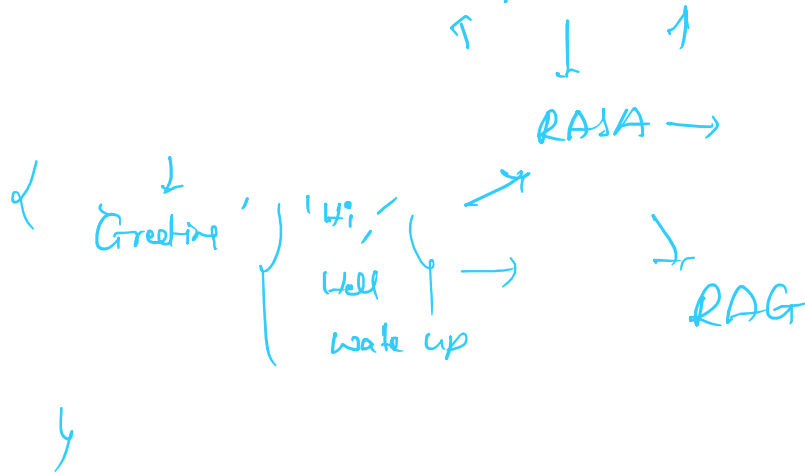
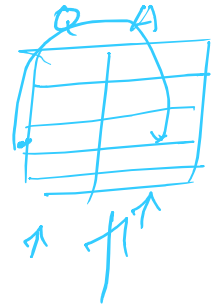
Applications of HLP:

1. Email spam detection.
2. Sentiment Analysis
3. NER (Named Entity Recognition)



Chatbot:

1. Generative Chatbot
2. Intelligent/Interactive Chatbot



- Google Translator
- Machine Translator / Lang Translator.

80% of internet data → text

LLM

Ambiguity.

- Sarcasm ✓
- Accent

1. Let's eat grandma
2. Chicken is ready to eat.
3. Car hit the pole while it is moving.

Football

↓
Break the leg.

$\Downarrow$
 text $\rightarrow$ Number-
 $\downarrow$
 vectorization

$\hookrightarrow$ converting text $\rightarrow$ num.

ML
 $\downarrow$
 Gender Color
 M ✓
 F ✓

- 0, 1, 2, ? ...
 $\uparrow$
 Basic lang repr.
- 1. CountVectorizer
 - 2. BoW (Bag of words)
 - 3. TF-IDF (Term Frequency Inverse Document Frequency)
- []
 $\uparrow$
 Distributed lang repr.
- 4. Word2Vec
 - 5. GloVe (Global Vectorizer for text embedding)
 - 6. FastText
- Contextual lang repr.
- 7. ELMo
 - 8. BERT
 - 9. GPT.

$\uparrow$
he is ...

Vectorization Vs Embedding:

- Vectorization will just convert any text data in some numerical repr.
- This vector can be sparse or dense repr.
- But high dimensional.
- Where as embedding is a technique to convert text to numbers with more advanced & sophisticated way.
- which capture semantic meaning for each word.
- It is low dimensional representation.

Text cleaning steps:

1. Convert the text into lowercase.

Python python PYTHON.

2. Tokenization.

↳ converting into words/sentence.

3. Remove spl char/punctuations.

4. Remove stop words.

↳ most repeated words.

5. pos tagging

6. Stemming

↳ convert each word into its root word.

studying studied study

study → Dim Red.

Lemmaization

↳ converting each word into its dictionary word.

studied, studied $\left\{ \begin{array}{l} \text{stem} \rightarrow \text{stud}^? \rightarrow \text{cut the last suffix.} \\ \text{Lemna} \rightarrow \text{study.} \end{array} \right.$

7. Vectorization.

real_news → 1

fake_news → 0

text	label
—	1
—	1
—	0
—	1
—	0

fake_news → 0

←	0
	1
	0

data = []

```
for i in real_news:  
    data.append([i, 1])
```

```
for j in fake_news:  
    data.append([j, 0])
```

```
pd.DataFrame(data,  
              columns=["text",  
                      "label"])
```

df["text"][0].lower()

"Government announces NEW education policy!!! #education 😊"

urls = "http \| S + / www \| S + "

re.sub(urls, " ", text)

re.sub("# \| w +", " ", text)

re.sub(puc, " ", text)

```
import string  
puc = string.punctuation
```

16 x 00

16 x 7F

16 x 0 = 0

16 x 7 = 112 + F

= 112 + 15 = 127

[^a-zA-Z\s]

0-9

10 A

11 B

12 C

13 D

14 E

15 F